Olympics' Herculean tech feats

The Summer Olympics have always been a refresher course to encounter new sports and marvel at athletes' superhuman, chiseled physiques on a quadrennial basis. This was before I knew that something like the Winter Olympics existed, of course. But soon enough, I was enamored by sights and scenes of snowcapped resorts, with plenty of skis, skateboards and matching apparels and too many picture-perfect postcard moments to keep track off. It was a bit hard to imagine and thus, something like the Winter Games held at Sochi, Russia, seemed to be nothing more than the greatest paid vacation for athletes from all over the world – but of course, that's not the case. The competition's as serious as it can be and there're so many events to track.

Not only do you get introduced to sporting disciplines you never knew existed, but also to technologies that help adjudicate, connect, and literally run the show. It's a massive undertaking, figuring out the tech to conduct and operate an event of the scale which experts are calling the most technologically advanced ever – with tens of thousands of athletes, support staff, organizers and the general public in attendance. Take for instance the network deployed by Avaya which provides a 54 Tbps backbone shared by four data centres, two media centres and 11 sporting venues. Internet providers at Sochi were transporting between 1-2 TB of data to and fro every single day.

Photography broke new ground at the Sochi olympics, too, given Getty Images' exhibits at the show. Apart from top-notch 360-degree panoramic images of every sports venue, remotely triggered photogrpahic cameras and even prototype cameras from the major manufacturers were used. All this to bring you fantastic images and photo-essays to make a lasting impression.

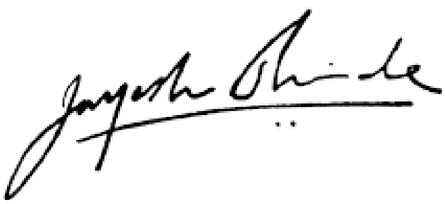
High-speed motion photography's great strides are also evident in Omega's official timekeeping apparatus installed at Sochi, and other prominent athletic events around the world. Preparations start two years in advance with camera installations and fibre optic cabling, leading to several months of testing the systems before showtime. Why? To catch differences of 0.003 seconds to separate the winner from second place, as it happened in a speed skate event. Mind boggling stuff.

Previously unseen security measures were deployed at Sochi as well. A system known as VibraImage "uses computer analytics of live video images to measure tiny muscle vibrations in the head and neck," according to NYT. It is designed to "detect someone who appears unremarkable but whose agitated mental state signals an imminent terrorist threat." Artec ID's Broadway 3D Face Recognition System at Sochi was "capable of identifying a person on the walk, it could also tell apart identical twins." There were military drones in the air, deep packet inspection of all incoming and outgoing communication from Sochi and more. What could possibly go wrong? It's Russia after all.

The athletes were also equipped with cutting edge tech. Take the kevlar and carbon fibre bobsled developed by BMW for Team USA, it's unlike anything you've ever seen. Speed skater Shani Davis' high-performance suit took more than two years to create in a research lab and is made from different fabrics including nylon, polyester, and spandex, to help minimize friction down to infinitesimal levels – all thanks to chemistry. There were Nike's padded boots with a futuristic cushioning layer developed by observing astronauts jump on the moon, vibration damping skis made of carbon nano-tubes, socks that were strong enough to be used in a bulletproof vests and so much more!

All these tech feats have helped me appreciate the nuances of this year's Winter Games. Thanks to the cutting edge behind-the-scenes tech and consumerization of smart gadgets, the athletes have never been more accessible, the coverage never more breathtaking, or the security never so water-tight. Of course, Putin and Russia had a lot to do with it – after all, they stockpiled several million cubic feet of snow for the Sochi games, according to reports, and deployed over 400 snow cannons to layer the slopes with manmade snow (just in case). But jest aside, tech deployed at these international sporting events has the potential to reveal more than what the annual CES, CeBIT or Computex have to showcase.

Lesson learnt: expect the unexpected at the FIFA World Cup later this year.



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